PingSheng Kao

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SUMMARY

Data Scientist with 6+ years of experience building measurement systems for multimodal, LLM, agentic, and production ML systems. Specializes in scalable model evaluation, ambiguous-ground-truth calibration, human-in-the-loop review, statistical risk analysis, failure discovery, and decision-ready monitoring.

EXPERIENCE

ML Data Scientist - Apple July 2020 - Present
Santa Clara, California

- Built simulated-user, agent-as-judge, targeted-scenario, and behavioral-metric workflows for an open-ended multimodal conversational agent, measuring intent misalignment and false inferred-state updates at scale.
- Scaled LLM- and agent-as-judge evaluation on an existing internal batch platform, resolving throughput and rate-limit issues, shipping versioned evaluator updates, and surfacing results in dashboards.
- Developed an auditable LLM evaluator deployed in an internal production workflow, combining expert-calibrated gold sets and human-in-the-loop validation; reached 90% alignment with experts on 40 examples.
- Led cross-functional translation from ambiguous user-research needs to engineering requirements and Data Operations labeling instructions, coordinating automated evaluation with human review.
- Designed Bayesian per-subject fairness analysis using Beta posteriors to estimate demographic-gap risk against a 10% product threshold.
- Built SQL/AWS Athena metric pipelines and leadership dashboards combining failure-pattern analysis, regression attribution, and online validation of offline-mined risks to guide development decisions.

RESEARCH CONTRIBUTION

GH-ESD: Grounded Hypothesis-Driven Error Slice Discovery for Instance-Level Vision Tasks 2025
Experiment contributor (credited as Sam Kao) - accepted to ECCV - arXiv:2512.24592

- Contributed experiments to ECCV-accepted research on grounded, hypothesis-driven error-slice discovery for instance-level vision tasks.

ADDITIONAL EXPERIENCE

Algorithm and Architecture Intern - Deep Learning Compression - Ambarella June 2019 - August 2019
Santa Clara, California

- Developed a saliency-based neural-network quantization heuristic that reduced an exponential search formulation to a linear-time method, and proposed a mixed-precision algorithm integrated into company software.

EDUCATION

M.S. in Computer Science and Engineering - University of Michigan, Ann Arbor 2018 - 2020
B.S. in Electronic Engineering - National Taiwan University 2013 - 2017
GPA: 3.9/4.0 (B.S.)

SKILLS

Programming & Data: Python, SQL, C++, JavaScript, AWS Athena, Tableau
AI Evaluation: Agent-as-Judge, LLM-as-Judge, User Simulation, Model Evaluation, DSPy
Statistics & Risk: Bayesian Analysis, Fairness Measurement, Experiment Design, Root-Cause Analysis
Human & Monitoring: Human-in-the-Loop Evaluation, Gold Sets, Failure Analysis, Dashboards, Evaluator Versioning